

Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

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Summarizing Random Variables

- Covariance
- Multivariate Distributions
- Law of Iterated Expectations (LIE)
- Law of Total Variance

Covariance

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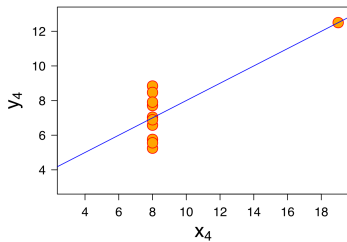
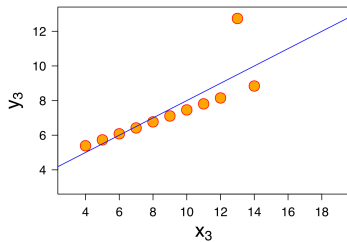
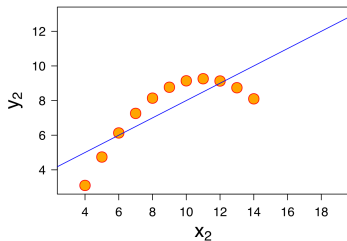
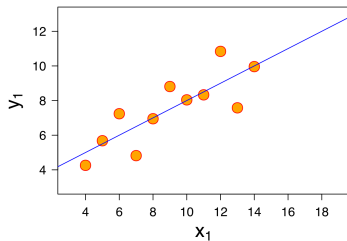
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Correlation= .816



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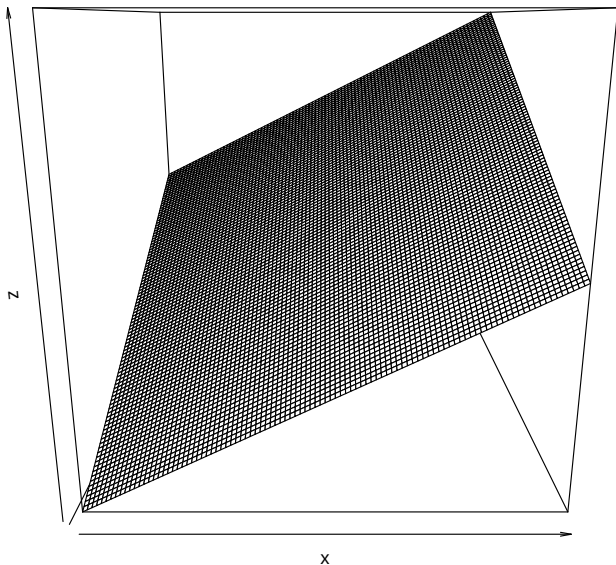
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Plot pdf $f(x, y) = x + y$

```
x <- seq(0, 1, 0.01)
y <- seq(0, 1, 0.01)
z <- outer(x, y, function(x, y){ x + y })
persp(x, y, z)
```



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$$\begin{aligned} E[X] &= \int_0^1 \int_0^1 x(x + y) dx dy \\ &= \frac{7}{12} \end{aligned}$$

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Conditional Expectation

$$\mathbb{E}(Y|X) = \int y \cdot f_{Y|X}(y|x) dy$$

$\mathbb{E}(Y|X)$ is the *best* way to predict Y given X .

Example of Joint Density

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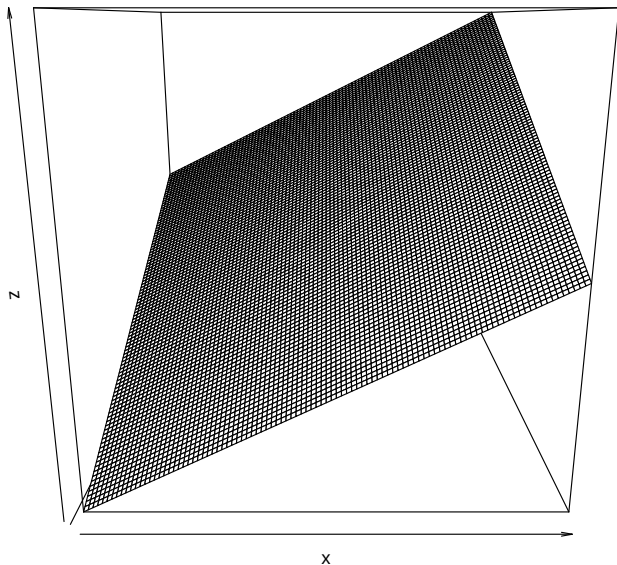
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$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{x + y}{\frac{1}{2} + y}$$



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Best Linear Predictor

Suppose we want a predictor $E(Y|X)$ that is linear:

$$f(X) = a + bX$$

Where a and b are set to minimize

$$E[(Y - (a + bX))^2]$$

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$$E[YX] - E[Y]E(X) = b[E(X^2) - E[X]^2]$$

$$b = \frac{E[YX] - E[Y]E(X)}{[E(X^2) - E[X]^2]} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

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Suppose X and Y have pdf $x + y$ for $x, y \in [0, 1]$.

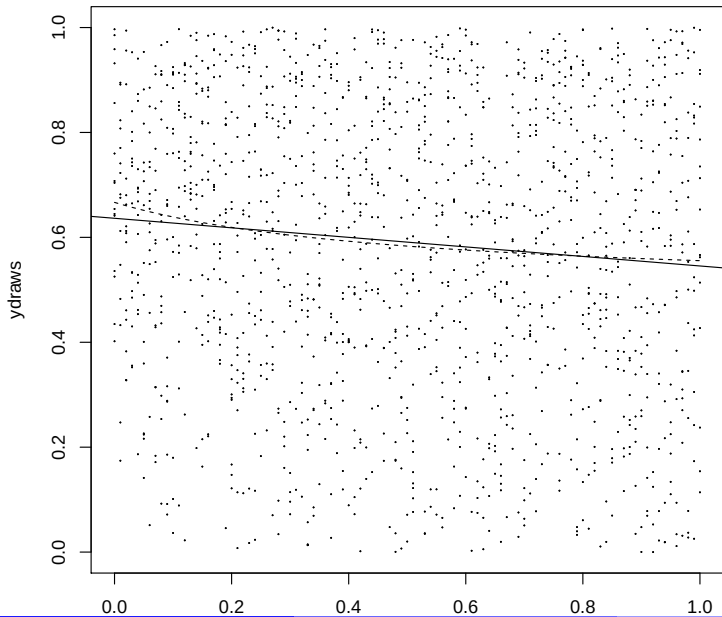
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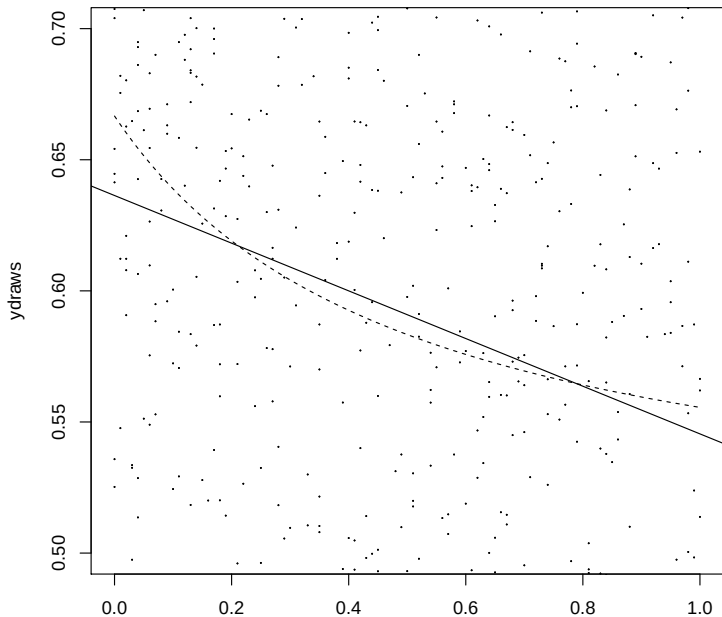
Example: $X + Y$

Suppose X and Y have pdf $x + y$ for $x, y \in [0, 1]$.

$$b = \frac{\text{cov}(X, Y)}{\text{Var}(X)} = -1/11$$

$$a = E(Y) - bE(X) = 84/132$$





Sums of Random Variables

Suppose we have a sequence of random variables X_i , $i = 1, 2, \dots, N$.
Suppose that they have joint pdf,

$$f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$$

- 1) $E[\sum_{i=1}^N X_i] = \sum_{i=1}^N E[X_i]$
- 2) $\text{var}(\sum_{i=1}^N X_i) = \sum_{i=1}^N \text{var}(X_i) + 2 \sum_{i < j} \text{cov}(X_i, X_j)$

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$$\text{var}(X_1 + X_2) = E[(X_1 + X_2)^2] - (E[X_1] + E[X_2])^2$$

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Consider two random variables, X_1 and X_2 . Then,

$$\begin{aligned}\text{var}(X_1 + X_2) &= E[(X_1 + X_2)^2] - (E[X_1] + E[X_2])^2 \\ &= E[X_1^2] + 2E[X_1X_2] + E[X_2^2] \\ &\quad - (E[X_1])^2 - 2E[X_1]E[X_2] - E[X_2]^2\end{aligned}$$

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Proof.

Consider two random variables, X_1 and X_2 . Then,

$$\begin{aligned}\text{var}(X_1 + X_2) &= E[(X_1 + X_2)^2] - (E[X_1] + E[X_2])^2 \\&= E[X_1^2] + 2E[X_1X_2] + E[X_2^2] \\&\quad - (E[X_1])^2 - 2E[X_1]E[X_2] - E[X_2]^2 \\&= \underbrace{E[X_1^2] - (E[X_1])^2}_{\text{var}(X_1)} + \underbrace{E[X_2^2] - E[X_2]^2}_{\text{var}(X_2)} \\&\quad + 2 \underbrace{(E[X_1X_2] - E[X_1]E[X_2])}_{\text{cov}(X_1, X_2)} \\&= \text{var}(X_1) + \text{var}(X_2) + 2\text{cov}(X_1, X_2)\end{aligned}$$



Definition

Suppose $\mathbf{X} = (X_1, X_2, \dots, X_N)$ is a vector of random variables. If $\mathbf{X}$ has pdf

$$f(\mathbf{x}) = (2\pi)^{-N/2} \det(\mathbf{\Sigma})^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \mathbf{\Sigma}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Then we will say $\mathbf{X}$ is a **Multivariate Normal** Distribution,

$$\mathbf{X} \sim \text{Multivariate Normal}(\boldsymbol{\mu}, \mathbf{\Sigma})$$

- **Regularly** used for likelihood, Bayesian, and other parametric inferences

Multivariate Normal Distribution

Consider the (bivariate) special case where $\boldsymbol{\mu} = (0, 0)$ and

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$f(x_1, x_2) = (2\pi)^{-2/2} 1^{-1/2} \exp \left(-\frac{1}{2} \left((\mathbf{x} - \mathbf{0})' \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} (\mathbf{x} - \mathbf{0}) \right) \right)$$

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$\rightsquigarrow$ product of univariate standard normally distributed random variables

Standard Multivariate Normal

Definition

Suppose $\mathbf{Z} = (Z_1, Z_2, \dots, Z_N)$ is

$$\mathbf{Z} \sim \text{Multivariate Normal}(\mathbf{0}, \mathbf{I}_N).$$

Then we will call $\mathbf{Z}$ the standard multivariate normal.

Properties of the Multivariate Normal Distribution

Suppose $\mathbf{X} = (X_1, X_2, \dots, X_N)$

$$\begin{aligned} E[\mathbf{X}] &= \boldsymbol{\mu} \\ \text{cov}(\mathbf{X}) &= \boldsymbol{\Sigma} \end{aligned}$$

So that,

$$\boldsymbol{\Sigma} = \begin{pmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_N) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) & \dots & \text{cov}(X_2, X_N) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_N, X_1) & \text{cov}(X_N, X_2) & \dots & \text{var}(X_N) \end{pmatrix}$$

Independence and Multivariate Normal

Proposition

Suppose X and Y are independent. Then

$$\text{cov}(X, Y) = 0$$

Proof.

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Then $\text{cov}(X, Y) = 0$.



Zero covariance does not **generally** imply Independent

Suppose $X \in \{-1, 1\}$ with $P(X = 1) = P(X = -1) = 1/2$.

Suppose $Y \in \{-1, 0, 1\}$ with $Y = 0$ if $X = -1$ and $P(Y = 1) = P(Y = -1)$ if $X = 1$.

$$E[XY] = \sum_{i \in \{-1, 1\}} \sum_{j \in \{-1, 0, 1\}} ijP(X = i, Y = j)$$

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Iterated Expectations (LIE)

Proposition

Suppose X and Y are random variables. Then

$$E[X] = E[E[X|Y]]$$

- Inner Expectation is $E[X|Y] = \int_{-\infty}^{\infty} xf_{X|Y}(x|y)dx$.
- Outer expectation is over y .
- This is analogous to the law of total probability.

Iterated Expectations

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Proof.

$$E[E[X|Y]] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X|Y}(x|y) f_Y(y) dx dy$$

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Example 1: LIE and Proxy Fighting

- Suppose the US is seeking a proxy partner for its fight against the government of Syria. Local allies come in three equally probable kinds, “tough”, “average” and “weak”.
- The US would be willing to give \$11,000 per fighter if the group is strong, \$10,000 if the group is average, and \$6,000 if they are weak.

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- $E[\text{price}_{\text{proxy}}] = \frac{1}{3} * (10k) + \frac{1}{3} * (6k) + \frac{1}{3} * (5k) = 7,000$
- If neither group knows their type, then they will be able to sell their services.

Example 1 continued: LIE and Adverse Selection

- Suppose that the proxy knows their type but the US does not.

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Example 1 continued: LIE and Adverse Selection

- Suppose that the proxy knows their type but the US does not.
- If the group proposes less than 6000, the US will think it is weak, and pay 6000.
- If the group were strong, they would want 10,000, but the US would only pay 9,000.
- Therefore tough proxies will not offer their services. This means that the US would pay at most:

$$E[\text{price}_{USA}] = \frac{1}{2}10,000 + \frac{1}{2}6000 = 8,000$$

Example 2: LIE and the Beta-binomial distribution

- Suppose we know our data is the result of a sum of binary outcomes, we would use the Binomial distribution.
- But the binomial assumes that the likelihood of success is a fixed value. What if it varies?
 - Heterogeneity: Not everyone has the same probability of voting for Trump.
 - Priors: Perhaps we have prior information about the value of p .
- Bayesian statistics begins with this idea, that the parameters of the model are random.

Beta Distribution

Definition

Suppose Y is a continuous random variable with $Y \in [0, 1]$ and pdf of Y given by

$$f(y) = \frac{1}{\mathbb{B}(\alpha, \beta)} y^{\alpha-1} (1-y)^{\beta-1}$$

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Then we will say Y is a **Beta** distribution with parameters α and β .
Equivalently,

$$Y \sim \text{Beta}(\alpha, \beta)$$

- Beta is a distribution on **proportions**
- $\mathbb{B}(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
- $E[Y] = \frac{\alpha}{\alpha+\beta}$

Example 2: LIE and the Beta-binomial distribution

Suppose

$$\pi \sim \text{Beta}(\alpha, \beta)$$

Example 2: LIE and the Beta-binomial distribution

Suppose

$$\begin{aligned}\pi &\sim \text{Beta}(\alpha, \beta) \\ Y|\pi, n &\sim \text{Binomial}(n, \pi)\end{aligned}$$

What is $E[Y]$?

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$$\begin{aligned}E[Y] &= E[E[Y|\pi]] \\ &= \int_{-\infty}^{\infty} \sum_{j=0}^N \binom{N}{j} j p(j|\pi) f(\pi) d\pi\end{aligned}$$

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Bayesian Inference

- $p(\pi|x, n) = \frac{p(x|\pi, n) * p(\pi)}{p(x)} \propto p(x|\pi, n) * p(\pi)$
- So we need a prior, $p(\pi)$, and a likelihood, $p(x|\pi, n)$.
- $p(\pi) = \text{Beta}(\alpha, \beta) = \frac{1}{\mathbb{B}(\alpha, \beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$
- Here α is the count of successes, β is the count of failures.
- $p(x|\pi, n) \propto \pi^x (1 - \pi)^{n-x}$

Bayesian Inference

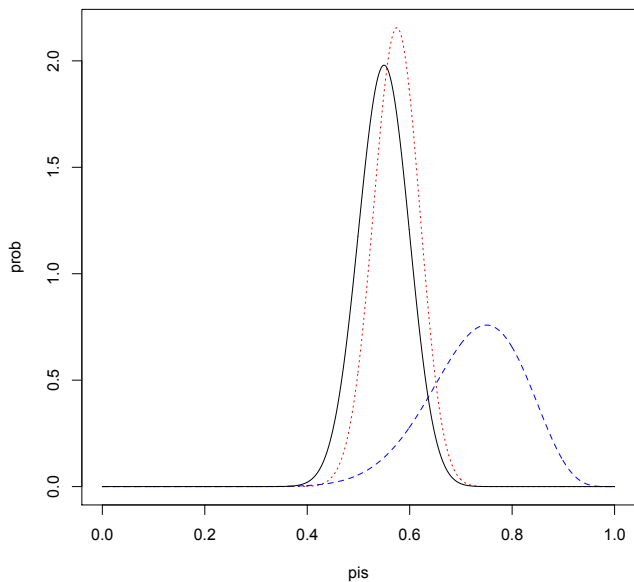
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- So we need a prior, $p(\pi)$, and a likelihood, $p(x|\pi, n)$.
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- Here α is the count of successes, β is the count of failures.
- $p(x|\pi, n) \propto \pi^x (1 - \pi)^{n-x}$
- $p(\pi|x, n, \alpha, \beta) \propto \pi^x (1 - \pi)^{n-x} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$
- $p(\pi|x, n, \alpha, \beta) \propto \pi^{x+\alpha-1} (1 - \pi)^{n-x+\beta-1}$

Bayesian Inference of Survey

- Suppose we did a pilot study of 20 people and found $\alpha = 15$ Trump supporters and $\beta = 5$ Biden supporters.
- We then take a survey of 100 people and find 55 Trump Supporters and 45 Biden supporters.
- $p(\pi|x, n, \alpha, \beta) \propto \pi^{15+55-1}(1 - \pi)^{5+45-1}$
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- Specifically, the Bayesian Estimate of π is Beta(Success+prior-1, Fail+prior-1)



Example 3: Conditional Expectation as an Estimator

Suppose Y is a random variable, and X is some data.

$$\hat{Y} = E[Y|X]$$

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This estimator's error centers on 0.

Correlation of Estimator and Estimation Error

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The error is uncorrelated with the estimate. This is what allows us to separate the signal from the noise.

Law of Total Variance

Suppose X and Y are random variables. Then

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Law of Total Variance: application

Suppose θ is a parameter, and x is data.

$$\text{var}[\theta] = E[\text{var}[\theta|x]] + \text{var}(E[\theta|x])$$

The expected variance of θ conditional on data (the posterior) is less than the variance in θ without the data (the prior).